AI For Accounting Games

Cycle 3 Report

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Submitted in partial fulfillment of the requirements for

COMP 4710 Senior Design

to the Department of Computer Science and Software Engineering,

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List of Figures **4**

System Metaphor **5**

Project Introduction **5**

Development **7**

Previous Development *7*

Cycle 1: 7

Cycle 2: 8

Current Cycle Work *9*

SSL Certificate 9

Single Sign On 9

Active Directory Parser 12

Role Based Access Control 13

Default Section 17

Logout 18

Data Clearing 19

Time Tracking 20

Security Improvements 20

Session Data Management 21

Final Testing 21

Future Work *22*

User Stories **23**

Design Documentation **27**

UML Diagram *27*

API Design *28*

Relational Database Diagram *30*

Management Plan **30**

Risk Mitigation & Management **32**

Lessons Learned **32**

Front End *33*

Back End *33*

Security *33*

Meeting Minutes **34**

March 24, 2026 @ 9am – Group Meeting *34*

March 31, 2026 @ 9am – Group Meeting *34*

April 1, 2026 @ 1pm – Group Meeting *34*

April 2, 2026 @ 9am – Capstone Presentation *34*

April 7, 2026 @ 9am – Group Meeting *35*

April 8, 2026 @ 8am – Group Meeting with Sponsor *35*

April 9, 2026 @ 2pm – Advisory Board Presentation *35*

April 14, 2026 @ 9am – Group Meeting *35*

April 16, 2026 @ 9am – Group Meeting with Sponsor *35*

April 19, 2026 @ 1pm – Group Meeting *36*

April 20, 2026 @ 8am – Group Meeting with Sponsor *36*

April 21, 2026 @ 9am – Group Meeting *36*

Appendix **37**

Jasson Buckman and IT Communications *37*

SSL Certificate Communications *38*

Advisory Board Presentation Invite *42*

Automated Population of Database *43*

Sponsor Approval **45**

List of Figures

Figure 1 Security Indication 9
Figure 2 Game Login Prompt 10
Figure 3 AUthenticate Page 11
Figure 4 Sarah Sessions Active Directory 12
Figure 5 Helen Grogan Active Directory 12
Figure 6 Helen Grogan logging in as a professor and using special permissions to be placed in section 001 12
Figure 7 Section Leaderboard 13
Figure 8 Global Leaderboard 14
Figure 9 Professor Dashboard Drop Down Menu 15
Figure 10 Professor Dashboard View 15
Figure 11 Admin Dashboard Select Section Menu 16
Figure 12 Admin and Professor Override 17
Figure 13Default Section Leaderboard 17
Figure 14: Logout Button within Settings Menu 18
Figure 15 Admin Clear Data Button 19
Figure 16 UML Diagram 28
Figure 17: API Diagram 29
Figure 18 Database Diagram 30
Figure 19 Workflow Management Diagram 31

System Metaphor

Cole Suchan

Our project is an educational gaming software designed to help students learn the fundamentals of accounting in a fun and interactive way. It is designed to help accounting students build a strong foundation in basic accounting principles by making what may be complex or confusing material more approachable. The goal is for students to be able to use this software as both a tool for learning and an engaging study aid.

The game is structured such that there are several different game modes that vary in both concept and difficulty, allowing students to begin with basic principles and work their way up to more advanced and involved topics. Gameplay also varies with each mode, allowing students to experiment with different forms of material consumption and keeping the game exciting.

The login process will be safe and intuitive, allowing students a simple way to access the game via their own personal account, which will in turn save and accumulate their stats each time they play. In addition to helping students, the game provides value to teachers too. Professors will be able to view gameplay data and statistics, allowing them to observe students' progress and identify areas in which their students may be struggling, and perhaps adjust their lessons accordingly.

Project Introduction

Jayden LaCross, Cole Suchan

Accounting Game is an ongoing collaboration with the Harbert College of Business and is being sponsored by Professor Andrew McLelland. The game is intended to aid students, specifically those in ACCT 2110 and 5110, to build and improve their understanding of the crucial accounting principles introduced and taught in these courses. Extractable data from game results will allow professors to track their students' progress, further enabling them to help their students in areas they may still be struggling with.

This application currently includes 3 games that cover basic accounting principles: *Debit vs Credit*, *The Five Elements*, and *Accounting Equation*. *Debit vs Credit* has users sort different accounting terms based on if they are a credit or a debit. *The Five Elements* is a similar game to *Debit vs Credit*, where students sort accounting terms into the five main accounting elements

(assets, expenses, liabilities, shareholder equity, and revenues). *Accounting Equation* is a quiz game that has students answer questions using the primary accounting equation (Assets = Liabilities + Shareholder Equity). The first two games function using the WASD or arrow keys to move a sprite that can pick up accounting terms and place them on a conveyer belt that directs it into the proper category. The space bar is utilized for picking up and placing down terms. The mouse and mouse clicks can also be used for controlling the sprite. *Accounting Equation* utilizes the mouse to select answer choices or equation values after the question is displayed.

Since starting this project in January, our group has achieved all of the goals we set out to accomplish this semester. Professor McLelland tasked us with two big jobs: OIT integration with SSO, and a user database to track in-game performance. With those goals came several challenges, but these objectives have been completed. *Accounting Game* now asks students, professors, and administrators to log in with AUthenticate when launching the game, and can accurately track stats, display names on the leaderboards, and show professors information, all without any hassle for the user. This creates a seamless gameplay and analytical experience for all of our clients at Auburn University, making A*ccounting Game* an extremely versatile edutainment tool for the Harbert College of Business.

Along with our main focus, several other improvements have been made to the game, such as visual improvements, enhanced feedback, customizable speed, settings menu, and admin/professor dashboards for student data. After completing our work this semester, our group is looking forward to handing this off to another team in the fall for even more improvements. We are also excited to see where Professor McLelland can go in terms of presenting this game at Business College conferences and potentially to other schools!

https://accounting-game.cse.eng.auburn.edu/

Development

Previous Development

Cole Suchan

Cycle 1:

This cycle focused on improving the game's usability, visuals, and gameplay experience, while laying out the groundwork for future features like login, data tracking, and SSO integration.

Key Accomplishments:

1. Gameplay Improvements
    a. Added a difficulty system for sorting-based games
    b. Converted an accounting quiz from multiple-choice to text input
    c. Updated/aligned question banks, updated game names per sponsor request
    d. Improved user feedback for answers
2. Visual/UI Improvements
    a. Increased resolution of game scenes
    b. Improved assets (e.g., business Aubie) & overall polish of various scenes/menus
    c. Settings menu -> Voume slider
3. SSO & Backend Preparation
    a. Met with Auburn IT to begin SSO (AUthenticate) integration planning
    b. Identified required SAML metadata and endpoints
    c. Established communication with SSO administrators

Challenges:

- Poorly documented legacy code
- Server access & VPN issues
- Difficulty accessing and updating the server (NGINX caching, SSH issues)
- SSO implementation delays due to external dependencies (IT coordination)

Cycle 2:

This cycle shifted toward backend integration and authentication, while continuing to enhance game engagement and data tracking capabilities.

Key Accomplishments:

1. Login & Data Tracking
    a. Built a test login system using localStorage (pre-SSO)
    b. Began tracking player stats (scores, time played, history)
    c. Updated leaderboard to use stored user data, expanded database schema
    d. Improved debugging feedback for login/testing
2. SSO (Single Sign-On) Progress
    a. Implemented SAML 2.0 setup using Auburn's AUthenticate system
    b. Created Service Provider (SP) configuration + certificates
    c. Deployed metadata endpoint for IT registration
    d. Requested key user attributes (name, email, role, course info)
3. Backend & Server Development
    a. Deployed a live FastAPI backend with API endpoints
    b. Configured Nginx routing to connect frontend ↔ backend
    c. Set up CORS policies for secure communication
    d. Created a systemd service to keep backend running reliably
    e. Updated routing to support /api/ endpoints
4. Gameplay Improvements
    a. Added a "streak system" to incentivize correct answers
    b. Continued work on feedback for incorrect answers
    c. Maintained improvements from Cycle 1 (difficulty, UI, etc.)

Challenges:

- SSO delays due to dependency on Auburn IT approval
- Complex issues with Nginx + FastAPI routing (404/405 errors)
- Dependency issues with SAML libraries (required extra installations)
- Accidental commits of sensitive certificates; required Git history fix

Current Cycle Work

SSL Certificate

Helen Grogan

One focus this cycle was increasing security on our website by moving our web application from HTTP to HTTPS. This was an important step to being able to utilize Auburn's Single Sign On (SSO). To do this we worked closely with Kelly Price to obtain an SSL certificate from Auburn (see emails in Appendix). Once the certificate was provided, we configured our nginx web server to handle HTTPS traffic on port 443 and redirect al HTTP traffic on port 80 to HTTPS automatically. The SSL certificate, issued through InCommon and Comodo CA, was placed on the server and verified by checking that the certificate and private key were a matching pair. After updating our nginx configuration to reference the certificate files, the site was successfully moved to HTTPS, giving it a secure connection.

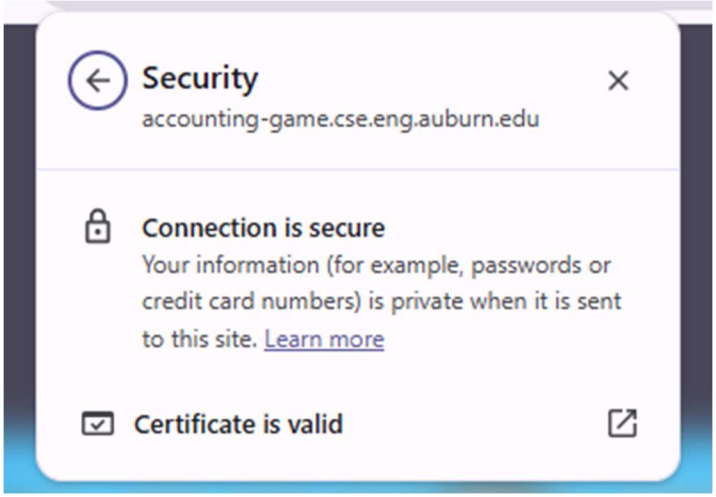

Figure 1 Security Indication

Single Sign On

Sarah Sessions and Helen Grogan

With HTTPS in place, we were able to move forward with implementing Auburn Single Sign On (SSO) using the SAML 2.0 protocol. This required coordination with Jason Buckman from Auburn's IT team to register our application as a Service Provider (SP) in Auburn's Identity Provider (IdP) system. We provided our SP metadata, available at https://accounting-

game.cse.eng.auburn.edu/saml/metadata, which Auburn IT used to configure the trust relationship between our application and their authentication system.

I coded a popup that prompts the user to use Auburn SSO to access our game. This popup comes automatically for each new use session meaning that users only have to login once per gameplay session. In other words, users will only have to sign in again if they close the tab they are playing on.

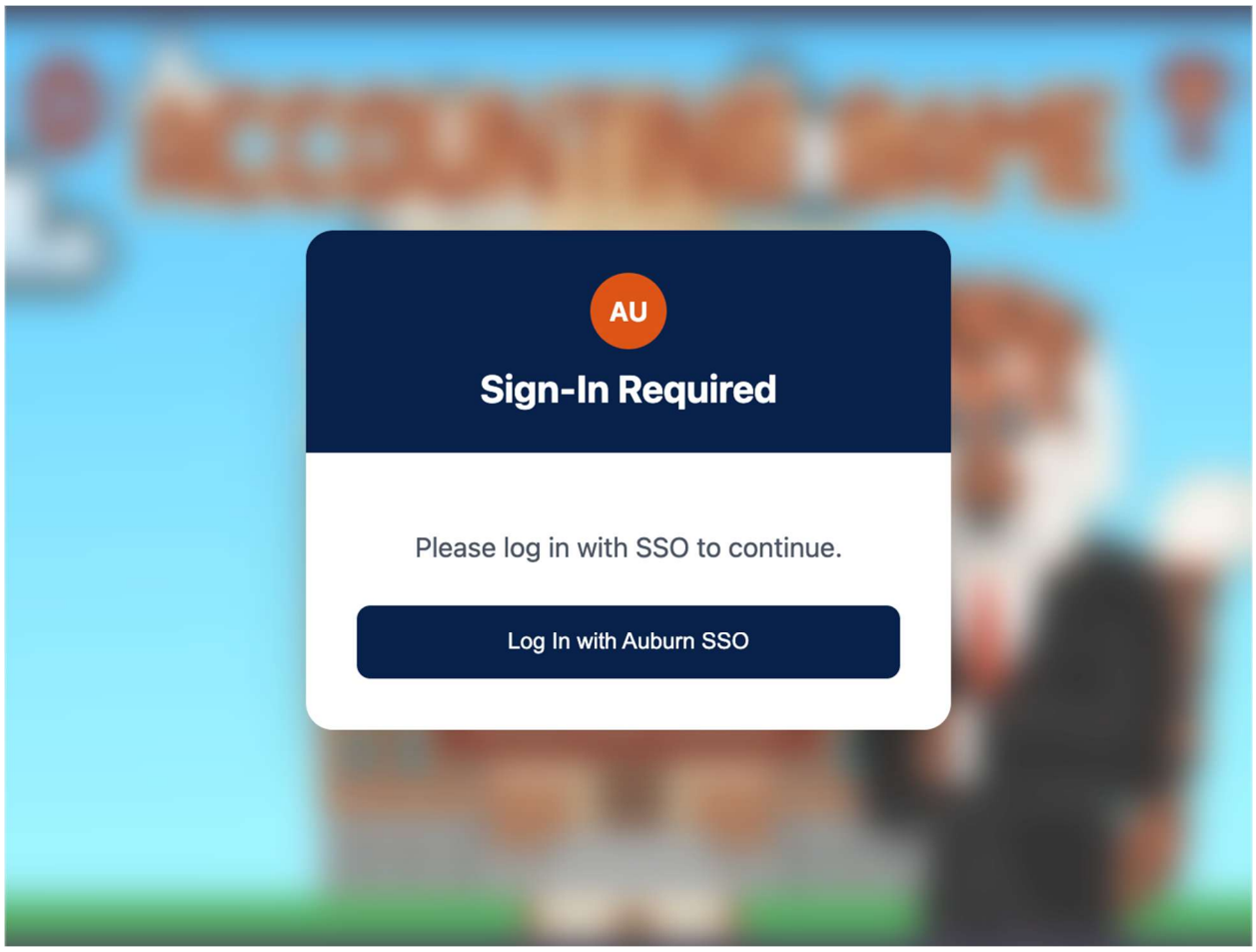

Figure 2 Game Login Prompt

On the backend, we integrated the python3-saml library into our FastAPI application to handle the SAML authentication flow. When a student visits our web application and licks login, they are redirected to Auburn's login page where they can sign in with their Auburn credentials.

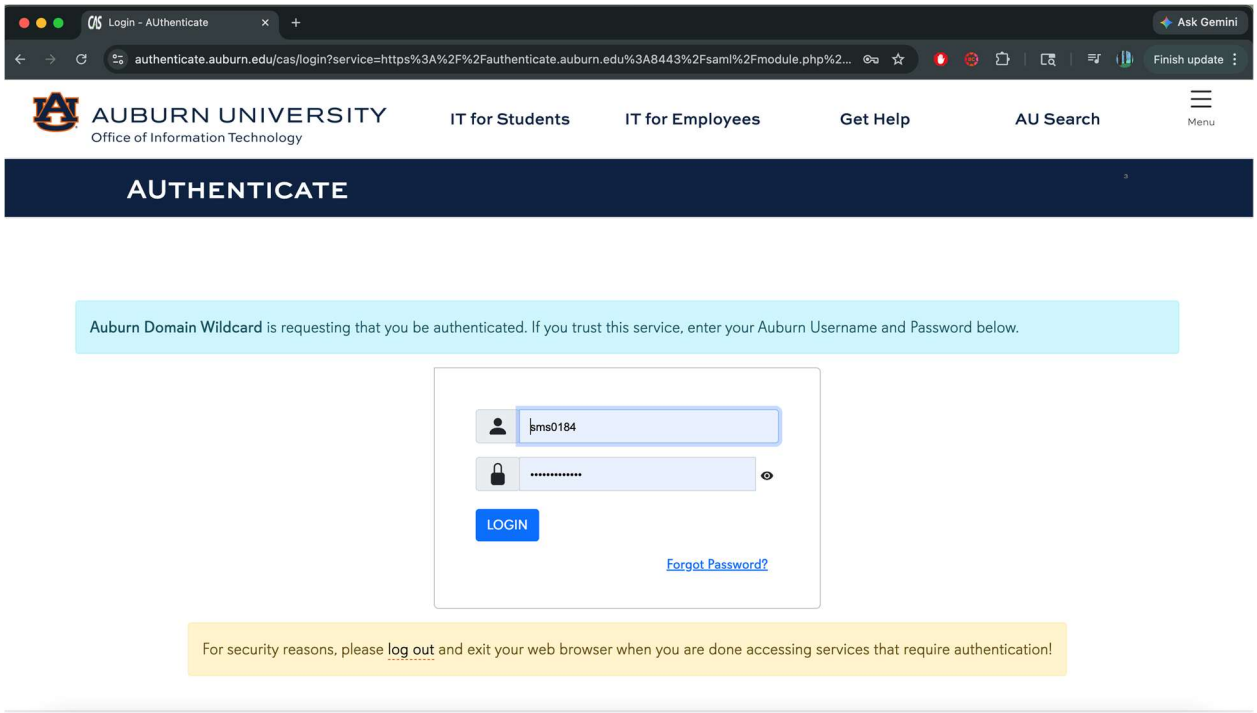

Figure 3 AUthenticate Page

Auburn's IdP then sends a SAML response back to our application's Assertion Consumer Service endpoint, where we validate the response and extract user attributes. This process gives us an Active Directory output which can be parsed and used to sort users for the role-based access control we have been testing and implementing. From this we can gather information such as first and last name, username, classes enrolled, university employment status, and planned graduation year. This information is then reformatted and passed into the preexisting logic. I had previously created a mock login process which gathered much of the same information directly from the user so that once IT got us what we needed for SSO integration we could quickly and easily utilize the information.

This has been the main goal of our work this semester and we are thrilled to have it finalized! By working with Auburn IT we were able to complete a metadata handshake and SAML integration which has been an invaluable real-world experience.

Active Directory Parser

Edward Calinsky and Sarah Sessions

To connect the newly integrated Single Sign on an Active Directory Parser had to be created, tested, and implemented. The AD information had to be taken from the hand off and turned into something usable for our data flow and for the information output our sponsor requested. As shown below, SSO provides a lot of user information which is not entirely useful for our purposes.

```
[redacted] 2026-04-18 19:55:33,443 - INFO - SAML Login successful for sms0184@auburn.edu.
 Professor: False. Sections: HR_stuemplist, 2026S_COMP6830_001, 2026S_COMP4710_001, 2026S_COMP5630_001, 2026S_COMP4810_001, 202
6S_COMP5120_002, 2026S_ANTH3010_EA1, 2026S_UNIV4AA0_UN1, 2026S_HONR2087_001, 2025F_HONR2087_EA4, 2025F_COMP5600_001, 2025F_COMP
6370_001, 2025F_COMP4730_001, 2025F_COMP6350_001, 2025F_ANTH3500_EA1, [redacted] ANTH_Minors, AIE
N_Majors, mallardaccess, ugrad, BoxUsers, AU_Active_Enrolled2, AU_GameNet2, lib_circ_students, CoE_Users, hr_austuemp_list, HON
R_Last_Term, HONR_Current_Term, ANTH_Current_Term, UNIV_Current_Term, CreativeCloudSubscribers_students, ANTH_Last_Term, COMP_L
ast_Term, AU_Active_Enrolled, AU_GameNet, CSCI_Majors, SR, OITLabs_Comp, OITLabs_Print_SO, waccess, OITLibrary_SO, 152_All_Gids
, 152_All, AUA_Employee_Gids, AUA_Gids, OITLABS_Print, [redacted] AU_Student_Gids, AU_Employee_Gids, COMP_Curren
lines 1-13/30 47%
```

Figure 4 Sarah Sessions Active Directory

```
[redacted] INFO:     10.99.16.239:0 - "GET /saml/login HTTP/1.0" 307 Temporary Redirect
[redacted] 2026-04-16 16:00:15,837 - INFO - SAML Login successful for hkg0017@auburn.edu.
 Professor: False. Sections: 2026S_COMP4710_001, 2026S_COMP5710_001, 2026S_COMP5700_001, 2026S_COMP4300_001, 2026S_COMP4810_001
, 2026S_UNIV4AA0_UN1, 2025F_COOP4920_002, [redacted] HBUS_Minors, [redacted]s, mallardaccess, ugrad, BoxUsers,
 AU_Active_Enrolled2, AU_GameNet2, SENG_Majors, CoE_Users, UNIV_Current_Term, COOP_Last_Term, CreativeCloudSubscribers_students
, AU_Active_Enrolled, AU_GameNet, SR, OITLabs_Comp, OITLabs_Print_SO, waccess, OITLibrary_SO, 128_All, 128_All_Gids, AUA_Employ
ee_Gids, AUA_Gids, OITLABS_Print, [redacted], AU_Student_Gids, AU_Employee_Gids, COMP_Current_Term, CoE_User, AU_
Gids
```

Figure 5 Helen Grogan Active Directory

This information is then processed by the parser and reformatted to fit our pre-existing data flow from our fake login process we produced last cycle. The information needed was the first and last name of the given user, their auburn username/email, if they are enrolled in ACCT 2110 or ACCT 5110 and what section. Additionally, we can see if a user is a professor and what class they teach so we can link a professor to their students. Finally, we can special permissions to our sponsor, his TA, and ourselves of Admin, as discussed in other sections of this report. This role flag is saved in the browser for each session in the same fashion we save a username between scenes so that users are not required to manually enter their credentials each time they change scenes.

```
- INFO - Professor 'hkg0017' has section override: 001
- INFO - User 'hkg0017' authenticated with role 'professor' and saved sections: 001
- INFO - Database entry created/updated for user 'hkg0017' with name 'Helen Grogan' and sections: 001
```

Figure 6 Helen Grogan logging in as a professor and using special permissions to be placed in section 001

Role Based Access Control

Sarah Sessions and Edward Calinsky

Our game has three roles each with their own permissions. The user with the least permissions is our default/student user. A student can see their own scores highlighted in their respective section leaderboard (if enrolled in ACCT 2110 or ACCT 5110) as well as the global leaderboard.

Figure 7 Section Leaderboard

Figure 8 Global Leaderboard

The next role is the professor role. This role is found as a flag in each user's active directory were we can see their instructor status and what classes they teach. If an induvial teaches multiple sections they are given a drop down menu to select which section they would like to view. They are shown information about their students sorted by student and game and provided with information about their top, bottom, and average scores in addition to their name and the time they spent on the game. Finally, we maintained the CSV button to allow instructors to download the data we present to them.

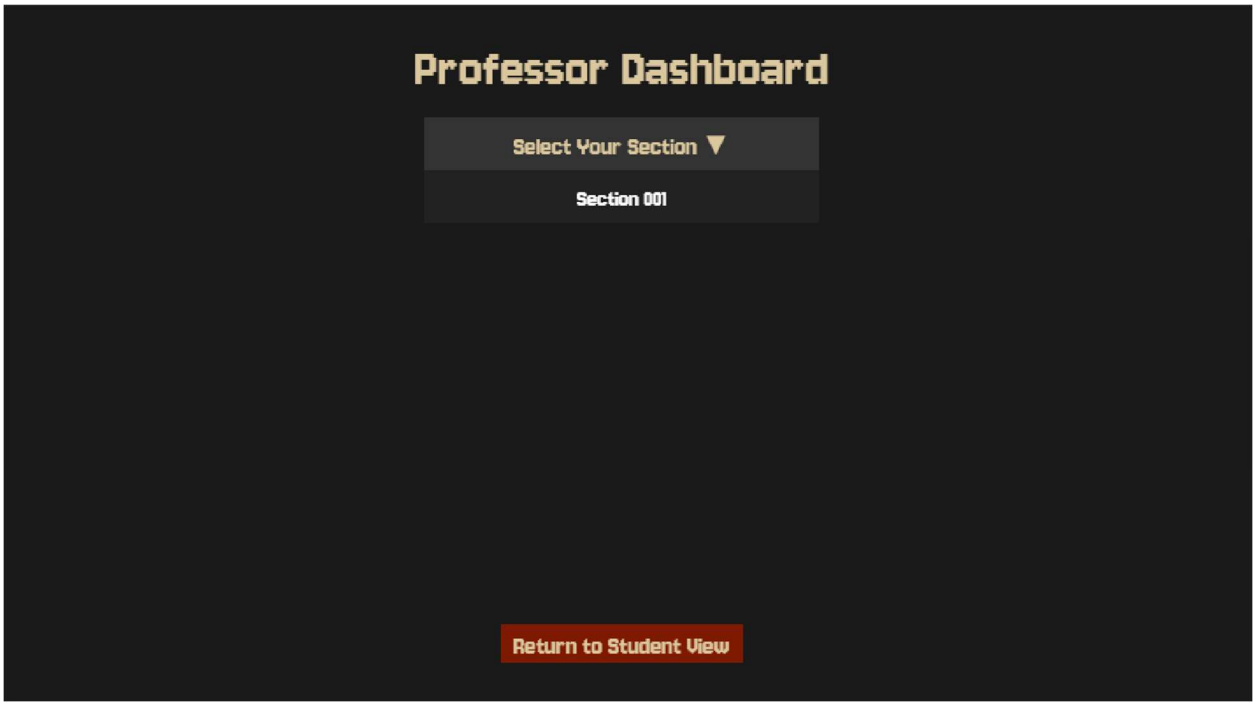

Figure 9 Professor Dashboard Drop Down Menu

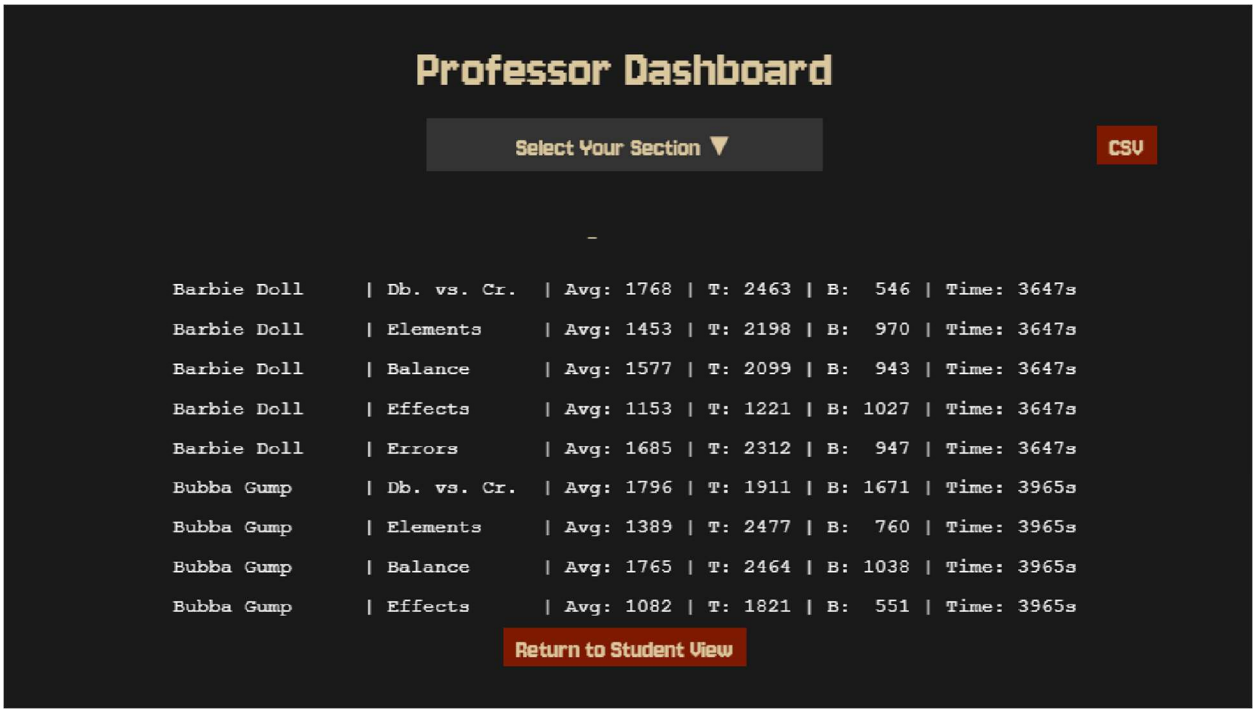

Figure 10 Professor Dashboard View

Finally, we have the admin role. This role is assigned through an additional excel sheet we have created to manage these permissions. The current admin users are only our sponsor, his TA, and current team members but this list of users can always be edited. The admin view searches the existing data base to display all unique sections in addition to showing all students and the global top scores. We have also implemented a data clearing button discussed in more detail later in this report.

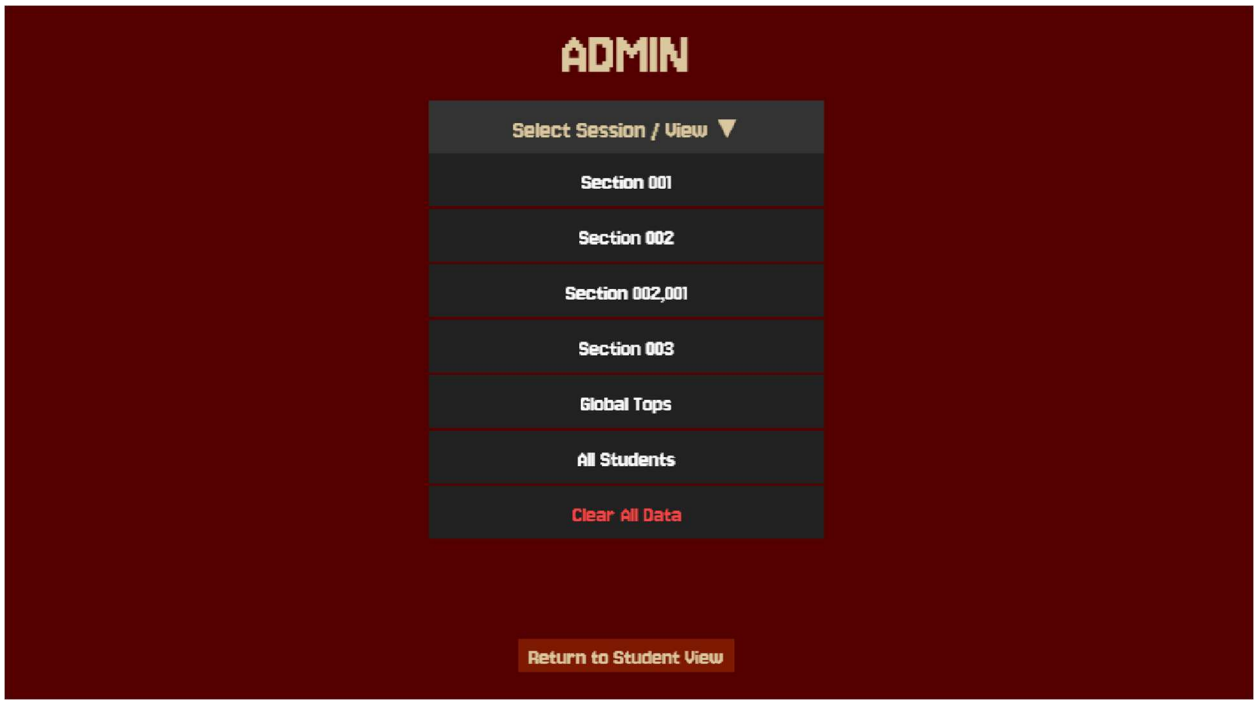

Figure 11 Admin Dashboard Select Section Menu

The above drop-down menu is a change from previous implementation to allow for an unknown number of sections as cleanly as possible. The section output for admin looks the same as the professor version but this view also has a list of all students who have played the game, regardless of if they are in a section or not, and a list of the top players.

Since we have made the admin role for our sponsor and his assistants, users can be added to this role using their Auburn username. For testing purposes, we also can assign the professor role. This allows team members to give themselves certain permissions to test functionality.

	A	B	C	D	E
1	**Admin (User ID)**		**TA (User ID)**	**Section (Year/Semester, Class Number, Section Number)**	
2	Example: **abc0123**		Example: **abc0123**	Example: **2026S_ACCT2110_001**	
3	sms0184		eac0124	002	
4	cds0122		hkg0017	001	
5					
6					
7					

Figure 12 Admin and Professor Override

Default Section

Helen Grogan

For users who are not enrolled in one of the designated accounting courses, we implemented a default section assignment for users whose Auburn Active Directory group membership does not include ACCT 2110 or ACCT 5110. When a user logs in via SSO and no matching course section is found in their AD groups, they are assigned section 'unknown' in the database. This includes professors who are not currently teaching one of the designated sections, ensuring they do not receive professor-level permissions.

Users assigned to the default section keep access to the game and are included in the global leaderboard, allowing them to play and compete across all game modes. However, when accessing the 'Section' tab on the leaderboard, they are shown the message 'MESSAGE HERE'. This ensures that the section leaderboard remains meaningful and only displays data for students actively enrolled in their specific course.

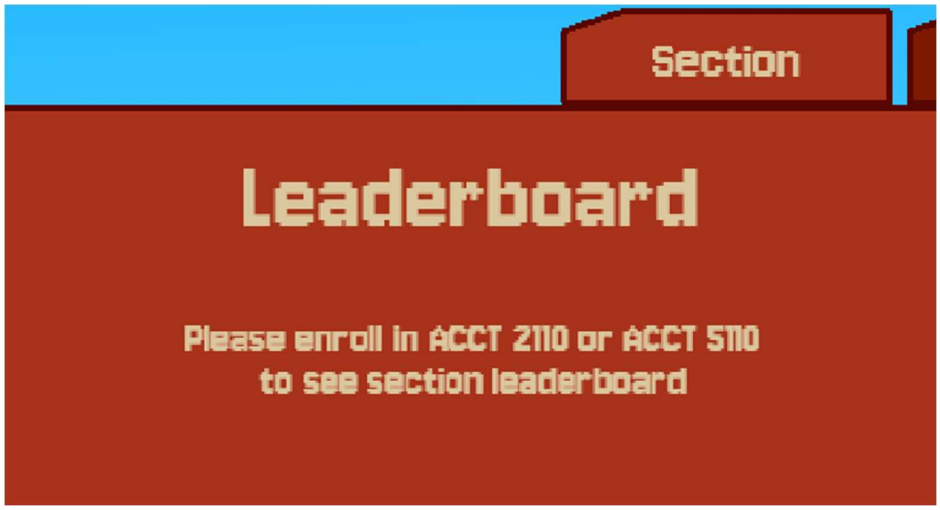

Figure 13Default Section Leaderboard

Logout

Cole Suchan

The settings menu was updated to replace a deprecated "Login Test" feature and role-selection buttons (Student, Professor, Admin) with a unified Logout button. This change reflects the transition from a mock authentication system (fake login) to a centralized Single Sign-On (SSO) workflow.

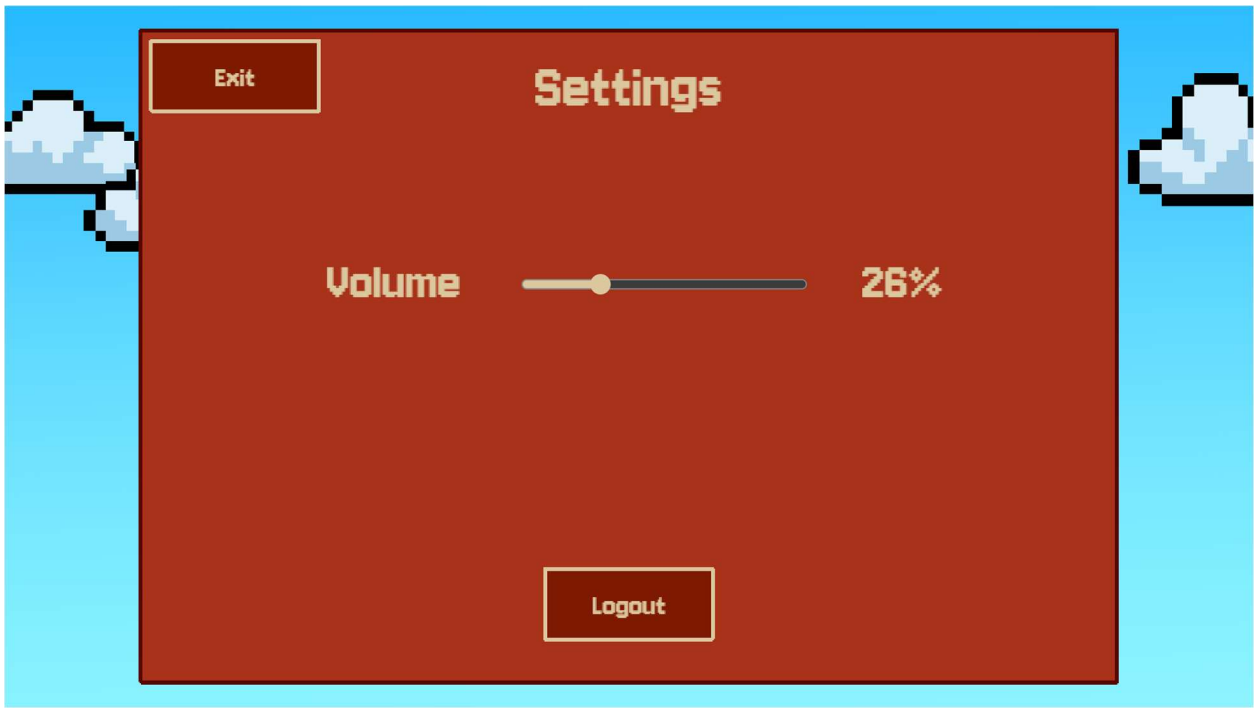

Figure 14: Logout Button within Settings Menu

The logout process performs three key actions:

1. Clears frontend authentication state by removing the `sso_completed` flag from `sessionStorage`, ensuring the application recognizes the user as logged out
2. Removes locally stored user data (e.g., username, section) from `localStorage` to prevent stale or incorrect user information from persisting
3. Redirects the browser to the SSO logout endpoint (`/saml/logout`), allowing the backend to properly terminate the user's authenticated session

This implementation ensures that upon logout, users are fully signed out both locally (local system is retired, purely for covering all bases) and through the SSO system. When the application reloads, the absence of the authentication flag triggers the login overlay, preventing further interaction until the user re-authenticates.

Data Clearing

Helen Grogan

To support end-of-semester data management, we implemented a clear data feature in the admin dashboard. This feature allows administrators to wipe all student game analytics and player profiles from the database at the end of the semester, providing a clean database for the next term. As student and professor section assignments are read fresh from Auburn's Active Directory each login, no additional data clearing is required.

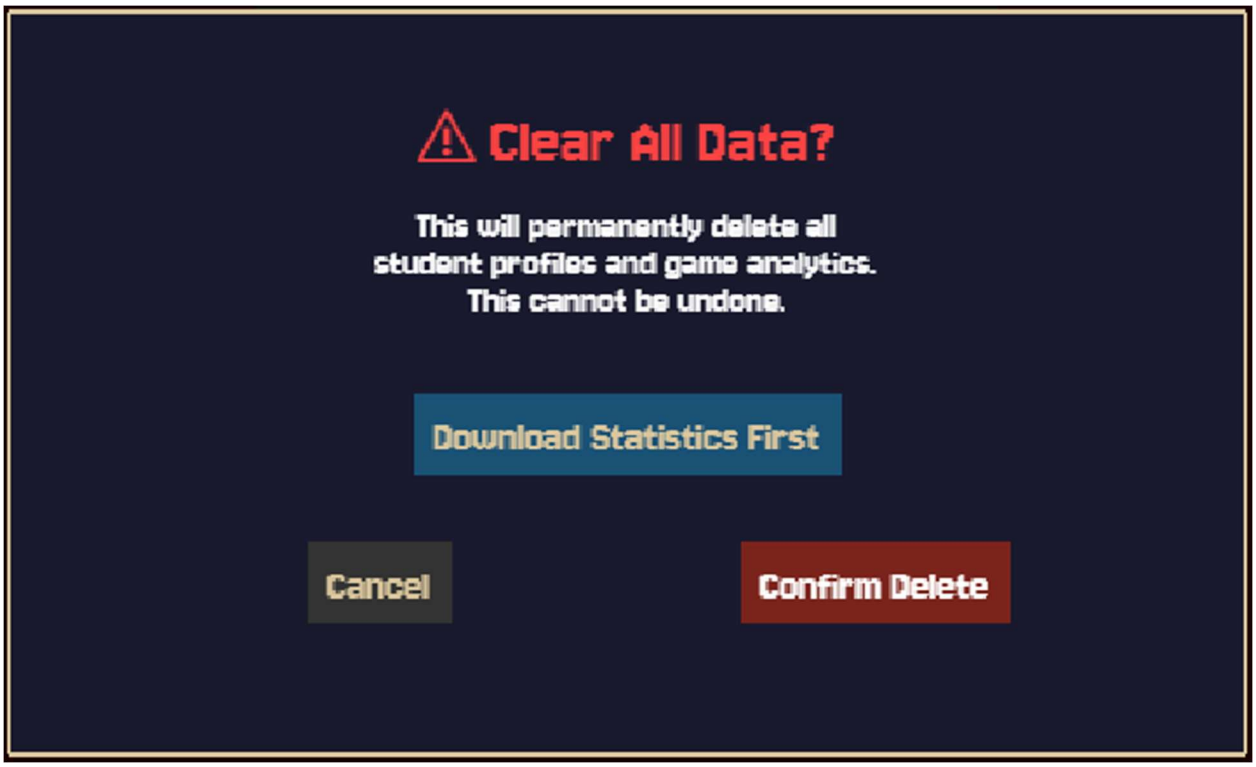

Figure 15 Admin Clear Data Button

This feature required implementation of a backend DELETE endpoint and a frontend confirmation. When an administrator clicks the 'clear data' button, a popup appears warning that

this action is permanent and irreversible. This popup also provides an optio to download a full CSV export of all student statistics before proceeding, ensuring no data is lost without the administrator knowing. Upon confirmation, the endpoint executes a database TRUNCATE operation on the game analytics and player profiles tables, which immediately removes all records. The admin dashboard then refreshes to reflect its now empty state.

Time Tracking

Sarah Sessions

It was also requested of us to track the amount of time a user spends playing each game. To do this, the Phaser supported register feature was used. This helps us to save information between scenes and works well with our existing game play flow where the games start in a main scene and then progress to the selected mode after a countdown. The timer starts after the 3-2-1 countdown to game play and is stopped at the end of the game. To end a game, players either the player runs out of time or complete all questions, at which point the elapsed time is calculated (in seconds) and saved along with the score. For accuracies sake, the time is saved and displayed in total seconds and can be converted after exporting the CSV of selected data. Additionally, to keep as much related data together as possible the time is recorded for each game and added up for dashboard display.

Security Improvements

Edward Calinsky

When creating the active directory parser, we were initally going to use the design of the fake login function we made before the SAML was fully finished. The data that was collected was stored in the browser's localStorage. This was where we planned to store the user's role and credentials. However, we discovered that a malicious user can directly access and edit their localStorage variable, allowing them to impersonate someone with higher privileges. To solve this problem, we implemented session middleware which encrypts and signs the parsed data and saves it to the browser as a cookie. Every time the data is needed by the frontend, the browser send a request to the backend containing the saved cookie. The backend checks the signature and

compares it to the payload, ensuring that it has not been tampered with. It then returns the saved role or user id to the frontend.

Session Data Management

Edward Calinsky

When the data from Auburn's Single Sign On is sent to the backend, it gets parsed and saved in a cookie in the browser session. The cookie contains the logged-in user's username, role, and the class sections that they participate in. The cookie is created and saved using session middleware, which encodes the data in base 64 and saves it with a signature afterwards. The signature is generated using a unique session id and combining it with a secret password saved in the .env file to create a signature, which ensures the cookie is not tampered with. To access the rest of the information, the username is used as a key to access the user's entry in the database.

Final Testing

Jayden LaCross

To ensure valid functionality of our game before semester's end, we tested all of the many endpoints involving the new features we had been working on. First off, after a user completes SSO, their data must be completely parsed, and intake must reflect all applications of the student data within the game. Things like user ID (for username on the leaderboard), section ID, and their class schedule must be correctly analyzed to determine how to correctly display data and elements for the user. If the user is a professor or an administrator, that data must also be handled correctly. Student, professor and admin views had to be correctly displayed, as well as all other applicable aspects of the game scenes.

The leaderboard, as well as the administrator and professor data views, all had to be correctly displayed for each user. Student data, scores, and times, all had to be correctly captured and implemented for their respective views. In addition, the database they are connected to must also be correctly displaying and updating the data. Administrators are also able to download the data to a .csv file, as well as clearing the data upon the end of the semester. All of these brand

new endpoints for our game, as well as the several points in the backend that were also altered, all had to be synchronized and tested to ensure everything was working correctly.

Future Work

Jayden LaCross

Our group was able to achieve all of the goals set out for us this semester. Although the game is pretty put together to the sponsor's standards as-is, future groups will always be able to make improvements. Even though we no longer have any features our team is working towards, our sponsor has expressed interest in presenting *Accounting Game* to others in the business education field, as well as potentially launching this game in other universities. Future improvements may also include visual enhancements, such as upscaling the resolution or updating textures, gameplay improvements such as adding games or enhancing gameplay, and structural improvements, such as improving backend workflow and implementation with our virtual machine.

If this game is indeed taken to other universities, consideration will also have to be made as towards the hosting of the game. Currently, the game is hosted on an Auburn University machine, only accessible through AU Wi-Fi. To make this game playable for other students, host machines at other universities would need to be instated, or our host would have to be given the ability to connect to students at other campuses. Server demand and web traffic would also have to be considered.

User Stories

Sarah Sessions, Jayden LaCross, Helen Grogan, Cole Suchan, Edward Calinsky

1.0 Student Login
Summary: As a student, I need to be able to login to let the website track my progress
Description: When I start the program, I will be prompted to log in somehow, either with my AU credentials or with some other method of on-site recordkeeping. This method should NOT hold ANY sensitive user information, such as passwords. This login information will then be saved for future use, so I will not have to log in every time I want to play (reduce friction). Once logged in, my scores/professor will be tracked for data usage.
Planned Hours: • Total: 30 • This Week: 20
Actual Hours: • 20
Coders: Helen, Sarah
Testers: Helen, Sarah
Reviewers: Helen
Story Status: Finished

1.1 Logged-in player interaction
Summary: As a logged in player, I should see the leaderboard
Description: Upon completing a game, the top 100 scores for that game, along with their players' initials, will be gathered and shown to the current player. Their position on the leaderboard will also be shown. When selecting a game, they may also show this to the player, with a personal best and data for motivation for improvement. The players title will be pulled from their logged information
Planned Hours: • Total: 30 • This Week: 5
Actual Hours: • 5
Coders: Edward, Sarah, Jayden
Testers: Sarah
Reviewers: Helen
Story Status: Finished

1.2 Logged-in player interaction
Summary: As a logged-in player, the game needs to know what section I am in and who my professor is.
Description: Once access has been gained to AU authenticate, we will be able to process section information and sort students. The user will see their section and professor displayed on login.
Planned Hours: • Total: 25 • This Week: 15
Actual Hours: • 20
Coders: Edward, Sarah, Helen
Testers: Cole, Jayden
Reviewers: Helen
Story Status: Finished

1.3 Logged in player interaction
Summary: As a logged in player, I need to see my progress such as time played, top scores, and improvement metrics.
Description: On the menu screen of the program, there will be a button that will take players to a statistics screen. This screen will display data to the user such as number of times played, highest score, and lowest score. This screen will also contain a button that will direct back to the main menu.
Planned Hours: • Total: 15 • This Week: 5
Actual Hours: • 5
Coders: Sarah
Testers: Edward, Jayden
Reviewers: Helen, Sarah
Story Status: Finished

1.4 Logged in player interaction
Summary: As a user, I need the ability to logout of the game
Description: There will be a button to logout of account from the main menu/settings page
Planned Hours: • Total: 10 • This Week: 3
Actual Hours: • 3
Coders: Cole
Testers: Sarah
Reviewers: Helen
Story Status: Finished

2.0 Admin/Professor Login
Summary: As a professor, I need to be able to login and see my course section and student information.
Description: On the menu screen of the program, there will be a button that will take professors to a statistics screen. Here they can select the course they would like to view data on. After selecting the course, the user will be able to see course and student statistics. This means the screen will provide a list of students enrolled within the course, along with their Auburn email addresses. This screen will contain a button to direct back to the main menu.
Planned Hours: • Total: 20 • This Week: 5
Actual Hours: • N/A
Coders: Sarah, Helen
Testers: Jayden
Reviewers: Jayden
Story Status: Finished

2.1 Professor reports
Summary: As a professor, I need to be able to see student report cards.
Description: On the menu screen of the program, there will be a button that will take professors to a statistics screen. Here they can select the course they would like to view data on. After selecting the course, the user will be able to see overall course statistics (average high score, average low score, average improvement) and individual student data. Professors will also be able to extract data into an excel spreadsheet. This screen will contain a button to direct back to the main menu.
Planned Hours: • Total: 25 • This Week: 25
Actual Hours: • 25
Coders: Sarah
Testers: Cole
Reviewers: Helen
Story Status: Finished

4.7 Player interaction
Summary: As a user, I want the UI to be cohesive
Description: Similar assets/visuals across scenes will be uniform, UI aspects should match theme of game and look visually appealing
Planned Hours: • Total: 5 • This Week: 5
Actual Hours: • 5
Coders: Cole
Testers: Jayden
Reviewers: Helen
Story Status: Finished

Design Documentation

Helen Grogan

The Accounting Game is a browser-based web application designed to use interactive gameplay to reinforce fundamental accounting principles for students. This application features three game modes of varying difficulties, each targeting a different core accounting principle. This cycle's primary focus was completing the authentication and role management system as well as the user statistics system.

Every authenticated user is given a unique record in our PostgreSQL database, allowing game statistics to be tied to a real student identity. Score data, session averages, and time played are stored and can be accessed through a global or section leaderboard. Professors are able to view and export their section's data through their dashboard, allowing them to see how their students are progressing with the course material.

UML Diagram

Helen Grogan

The updated UML class diagram reflects the significant architectural changes made this cycle. The previous custom login system has been replaced with the SAML 2.0 authentication through Auburn's Identity Provider, represented by the SAMLAuth and AuburnIdP classes. Upon successful authentication, the system parses the user's Active Directory group membership to determine their role and assigns them to one of three role-based views: StudentView, ProfessorView, or AdminView. The LeaderboardMenu is the routing point for these assignments. Users without an active enrollment in ACCT 2110 or ACCT 5110 are assigned a default section and keep access to only the global leaderboard.

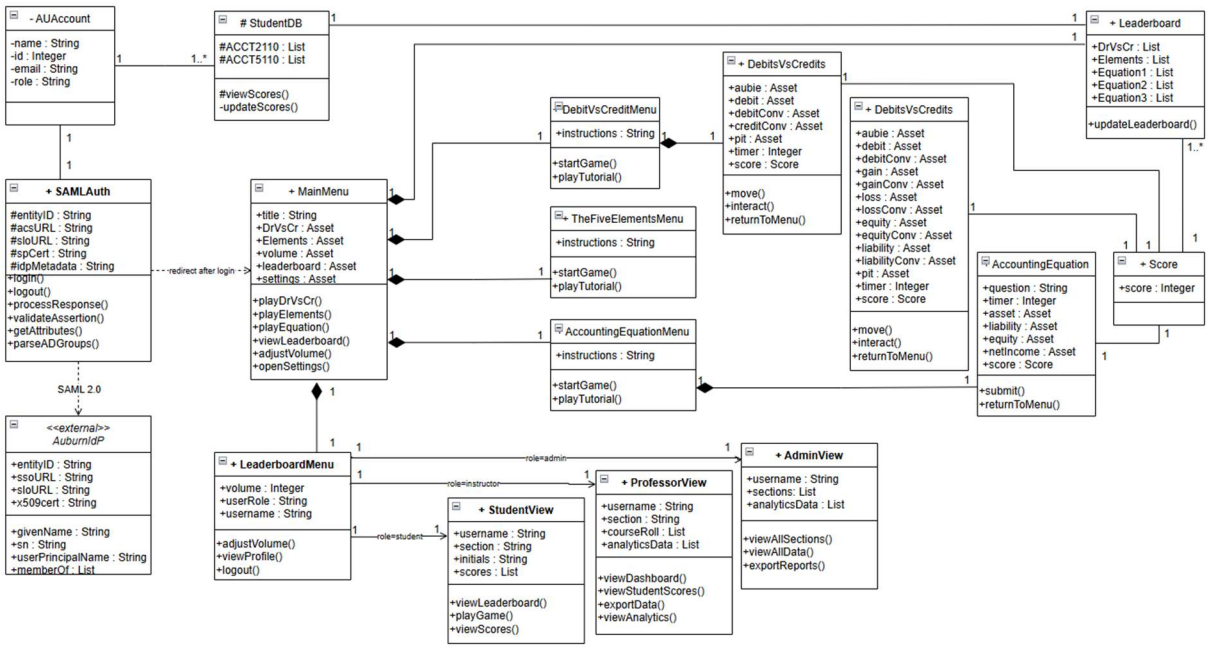

Figure 16 UML Diagram

API Design

Cole Suchan

At a high level, the system consists of a client interface, a web server, a FastAPI backend service, and a database, with an authentication layer controlling access between components.

The client represents the front-end application (Accounting Game). It communicates with two backend components:

- The web server, which handles front-end hosting and basic routing
- A FastAPI service, which acts as the main API layer responsible for processing application logic and handling data operations

Between the client and the FastAPI service is an authentication layer. This layer ensures that any requests sent from the client to the API are properly authenticated before being processed. We use Auburn's SSO service to authenticate users (mainly accounting students).

The FastAPI service sits between the client and the database and serves as the core backend:

- It receives authenticated requests from the client and processes related logic
- It communicates directly with the database to read or write data (leaderboard scores)

The database (bottom of the diagram) is responsible for storage of user data. It only interacts with the FastAPI service, not directly with the client or web server, which helps maintain the security of the game/users

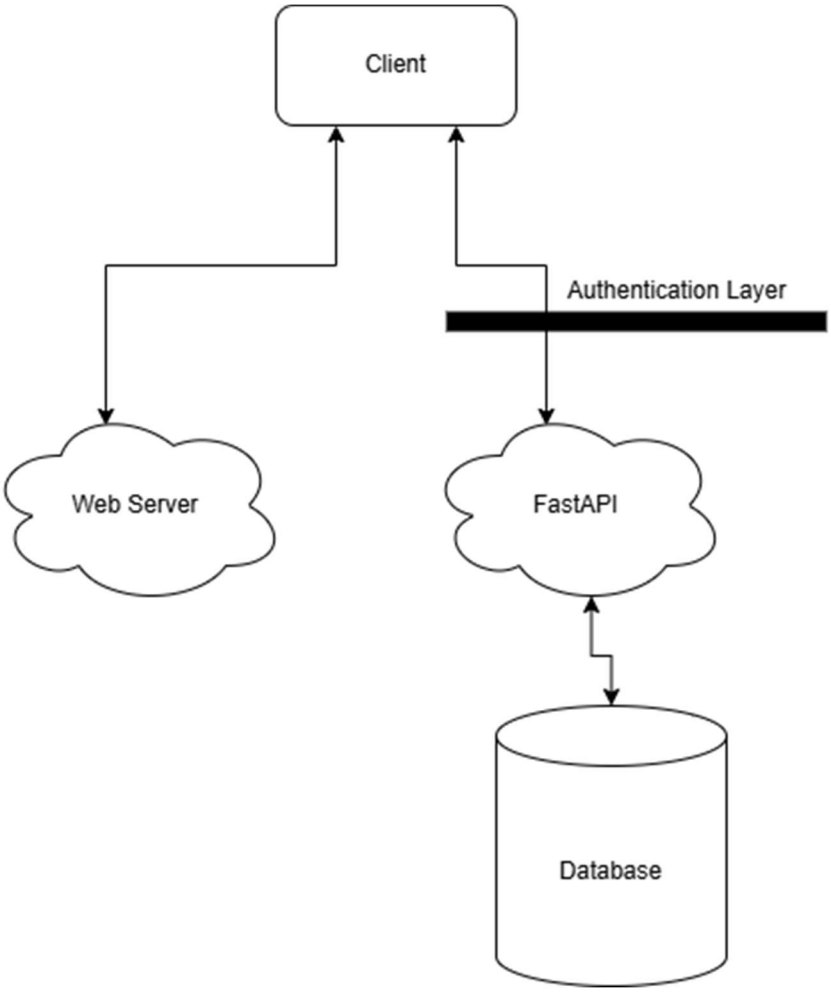

Figure 17: API Diagram

Relational Database Diagram

Sarah Sessions

Below is a crow's foot representation of our relational database. We utilize Postgres and PSQL to manage and interact with this database which is structured in the following fashion.

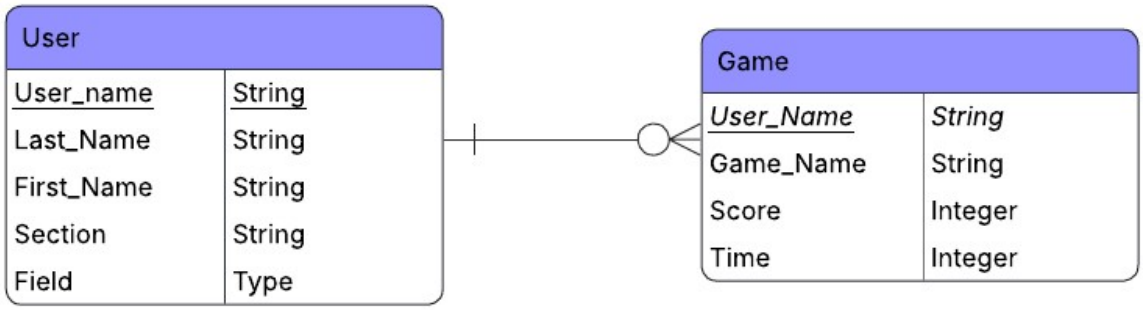

Figure 18 Database Diagram

We have user and game tables where each user can have zero or more game tables connected to them by their unique Auburn username. To have the more clean and accurate data possible we record every single game played and query (with pre-configured SQL) the database to produce the metrics seen in the various dashboard views.

Management Plan

Sarah Sessions

As we reach the end of this project, our management plan has continued to expand. This cycle had many large items for us to complete in a limited amount of time. For us to accomplish this in an efficient manner task delegation was extremely important to our workflow. This cycle saw lots of collaborative work on large and interconnected parts of our program which we took long hours to complete with the utmost care. As the following diagram shows, we have wrapped up our work for this semester by finally joining all of the separate pieces we have been developing into a finalized product.

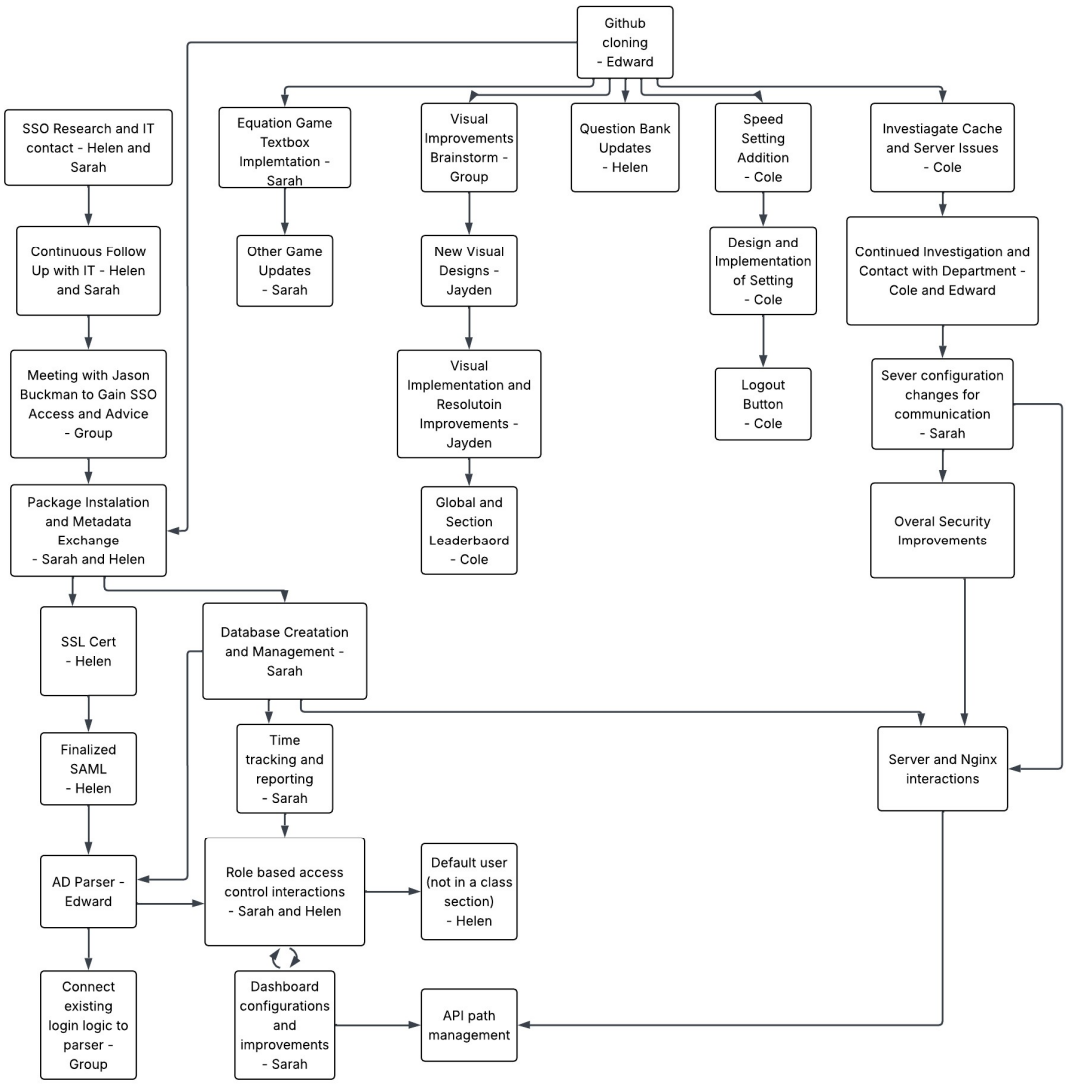

Figure 19 Workflow Management Diagram

Risk Mitigation & Management

Helen Grogan

Source Code: https://github.com/EdwardACaly/Accounting-Game.git

During development, version control is handled by GitHub, allowing full traceability of member commits. By enforcing proper branch management, we keep the stable production code isolated from code still in development. This approach minimizes the risk of merge conflicts and prevents unfinished or unverified code from impacting the live environment.

This application integrates with Auburn University's AUthenticate Single Sign-On system. This allows users to authenticate using their existing campus credentials, utilizing Auburn's security protocols and Multi-Factor Authentication for faculty permissions. To maintain FERPA compliance, the application implements Role-Based Access Control. By strictly separating student and faculty permissions, we ensure that sensitive student information is only accessible to authorized users, effectively mitigating the risk of data leaks and regulatory non-compliance.

Lessons Learned

Sarah Sessions and Edward Calinsky

This final cycle has given us the opportunity to work with a wide array and skills and tools in a full stack development environment. Our focus this semester has been bridging the gap between what the user sees on the front end and the functionality of the back end and making the website secure enough to be handed to students. We have done this this through a combination of software development, security, and academic tools including (but not limited to) Phaser, JavaScript, Vite, Node.js, NPM, Python, FastAPI, PostgreSQL, GitHub, and LaTeX. We also have put concerted effort into learning several development skills culminating in an understanding of state management, responsive scaling, relational mapping, database management, performance metric, dynamic data masking, input handling, role-based views, API integration, and solutions to many common issues that come up in web development. Each of these tools and techniques are powerful but come with their own challenges and learning curves which we have overcome this semester.

Front End

Sarah Sessions

A big lesson this semester has been that even features that seem “simple” upon first glance can be fragile and require careful testing and implementation every time. Things like monospace for tables can make initial creation easier but, in the end, old school methods like old school string padding can give better results! This shows that testing multiple routes is always best practice even when we think we’ve found a quick and easy solution. We also learned to always be willing to walk something back if it isn’t working. There were many times in many areas of the code that someone’s first attempt just didn’t work out but by working together we were always able to find a better solution.

Back End

Sarah Sessions and Edward Calinsky

The back end is where the magic happens, but it is also where the most frustrating bugs live. API paths and Phaser are extremely picky about formatting, and a single character difference can send a developer down a rabbit hole trying to find it. We also struggled with the way Nginx treats the endpoint paths that we organized in the backend, and whether parts of the paths would be stripped off or not. The URL generation functions were also inconsistent, causing us to send requests to endpoints that did not exist. These vulnerabilities were exposed when we fully implemented the use of live URLs and requests. Thankfully, we were able to use the console log to identify each mismatched path and correct it as we switched off of local testing.

Security

Edward Calinsky

A core security change we made this cycle is the transition from using local storage to cookies to track user data between scenes. This may have been a slightly harder version to implement and required changing previous code, but it is significantly safer for our users’ data. Using signed and encrypted cookies ensures that malicious users cannot tamper with their assigned roles and or impersonate a different user.

Meeting Minutes

Helen Grogan

March 24, 2026 @ 9am – Group Meeting

- Worked on Cycle 2 oral presentation
- Mapped out tasks for Cycle 3
- Delegated tasks for Cycle 3

Total Hours: 2

March 31, 2026 @ 9am – Group Meeting

- Worked on Capstone poster
- Fixed leaderboard issues
- Enhanced UML diagram

Total Hours: 2

April 1, 2026 @ 1pm – Group Meeting

- Finalized and printed poster
- Assigned presenting roles
- Practiced presentation

Total Hours: 3

April 2, 2026 @ 9am – Capstone Presentation

- Presented project at senior capstone expo
- Discussed project and industry with judges
- Learned about other senior design projects
- Listened to keynote speaker

Total Hours: 4

April 7, 2026 @ 9am – Group Meeting

- Delegated new cycle 3 tasks
- Discussed feature implementation
- Tested new implementations

Total Hours: 2

April 8, 2026 @ 8am – Group Meeting with Sponsor

- Demoed new feature implementations and discussed feedback
- Discussed security updates
- Confirmed action plan for professors with multiple sections

Total Hours: 1

April 9, 2026 @ 2pm – Advisory Board Presentation

- Presented capstone poster to Advisory Board
- Discussed industry with Advisory Board
- Discussed Auburn curriculum with Advisory Board

Total Hours: 1.5

April 14, 2026 @ 9am – Group Meeting

- Began Cycle 3 written report
- Mapped out all remaining fixes and features
- Discussed AD parser filtering

Total Hours: 2

April 16, 2026 @ 9am – Group Meeting with Sponsor

- Discussed bug in Game 3 Level 1
- Demoed SSO login process and other features

- Discussed feedback and end-to-end flow

April 19, 2026 @ 1pm – Group Meeting

- Finished role-based access control implementation
- Implemented a variety of bug fixes
- Handled various user permissions features
- Troubleshot and tested features

Total Hours: 14

April 20, 2026 @ 8am – Group Meeting with Sponsor

- Demoed full web application
- Discussed feedback
- Discussed small bugs and fixes

Total Hours: 1

April 21, 2026 @ 9am – Group Meeting

- Finished written report
- Worked on oral presentation/demo
- Worked on user/dev documentation

Total Hours: 2

Appendix

Sarah Sessions, Helen Grogan

Jasson Buckman and IT Communications

Helen Grogan

Reply Reply all Forward

To: Jason Buckman

Wed 4/8/2026 12:18 PM

Cc: Sarah Sessions

Hi Jason,

Hope you're doing well. I wanted to follow up on our SAML SSO integration. Since we last spoke, we have gotten HTTPS set up on our server and wanted to let you know that our metadata URL has changed.

Our updated metadata URL is:
https://accounting-game.cse.eng.auburn.edu/saml/metadata

The previous URL was using HTTP and included a port number, so the ACS and SLO endpoints have changed as well. All of the updated URLs are available in the metadata above if you need to update your registration on your end.

Could you update the registration with the new metadata URL and let us know once that's done? We'd like to test the full login flow as soon as possible.

Thank you again for all your help with this!

War Eagle!

Helen Grogan and Team
Undergraduate
Computer Science and Software Engineering
Auburn, AL 36849
Phone: 248.463.0383
hkg0017@auburn.edu

AUBURN UNIVERSITY
SAMUEL GINN
COLLEGE OF ENGINEERING

Jason Buckman

Reply Reply all Forward

To: Helen Grogan

Wed 4/8/2026 12:49 PM

Cc: Sarah Sessions

I've been expecting to hear from you. Glad you got that SSL part worked out. As it turns out, the original metadata you had was already assuming https on the default port for all of your URLs, so I didn't have to change anything. Let me know if your testing gives you errors.

Helen Grogan

Reply Reply all Forward

To: Jason Buckman

Wed 4/8/2026 1:09 PM

Cc: Sarah Sessions

Great news, we just successfully tested the full login flow and everything is working. Users are able to log in with their Auburn credentials and get redirected back to the game as expected.

Thank you so much for all your help getting this set up. We'll let you know if anything comes up during further testing.

Helen Grogan and Team
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Phone: 248.463.0383
hkg0017@auburn.edu

AUBURN UNIVERSITY
SAMUEL GINN
COLLEGE OF ENGINEERING

...

Reply Reply all Forward

SSL Certificate Communications

Helen Grogan

Reply Reply all Forward

To: itservicedesk@auburn.edu

Tue 3/31/2026 9:03 AM

Cc: Sarah Sessions; Jason Buckman

Good morning,

I'm a student working on the Accounting Game, a COMP 4710 Senior Design project at Auburn. We have a web application hosted on an Auburn VM at accounting-game.cse.eng.auburn.edu and we are trying to get SSL set up for our server.

We are in the process of implementing Auburn's SSO for student authentication and have already been in contact with Jason Buckman regarding the SAML configuration. The remaining blocker on our end is getting HTTPS set up, as Auburn's IdP requires a secure connection to complete the login flow.

Could you point us in the right direction for obtaining an SSL certificate for our server? We weren't sure if this is something Auburn IT handles for university-hosted VMs or if we need to set it up ourselves.

Any help or guidance would be really appreciated!

Helen Grogan '26
Undergraduate
Computer Science and Software Engineering
Auburn, AL 36849
Phone: 248.463.0383
hkg0017@auburn.edu

AUBURN UNIVERSITY
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Reply Reply all Forward

Summary: Complete to Route Properl... (RITM0155107)
Author: Savanna Spohn
Comment: "*New comment on TASK0180228**
- Hi Helen,

There is an SSL request form but it has to be filled out by an IT provider. We will send this over to Engineering IT.

Thank you,

Savanna Spohn
Endpoint Support Analyst, Student Supervisor
Auburn University
Office of Information Technology
3010 B RBD Library, 231 Mell Street, Auburn, AL 36849
O: 334.844.9391
sns0017@auburn.edu"

Summary: SSL Certificate Request (TASK0180228)
Author: Helen Grogan
Comment: "*New comment on RITM0155107**
- Good morning,

Thank you for your help with this. Would you please forward us that email and provide us with an engineering IT email address to contact about updates on our request?

Thank you so much for your assistance!

War Eagle!

Helen Grogan and Team"

KP

Kelly Price

To: Helen Grogan

You replied on Tue 4/7/2026 9:31 AM

Hi

I have submitted a certificate request to OIT and am waiting on them to send me the cert.

HG

Helen Grogan

To: Kelly Price

Hello,

Thank you so much for the update, let us know if there is anything you need form us.

War Eagle!

Helen Grogan '26

HG

Helen Grogan

To: Kelly Price

Reply Reply all Forward

Tue 4/7/2026 9:31 AM

Good morning,

Hope you are well. Are there any updates on the SSL certificate request? If not, is there an OIT email we can contact for updates or anything we can do to expedite this process?

Thank you so much for all of your time and assistance!

War Eagle!

Helen Grogan and Team

KP

Kelly Price

To: Helen Grogan

I'm checking on it...

Sorry for the delay.

Kelly Price

To: Helen Grogan

Reply Reply all Forwa

You replied on Tue 4/7/2026 5:12 PM

I have received the certificate and placed it in /etc/certificate on accounting-game. I haven't done anything more, as I am unsure what you need,

Helen Grogan

To: Kelly Price

Reply Reply all Forward

Tue 4/7/2026 4:34 PM

Thank you for getting that set up! We were able to locate the certificate files in /etc/certificate and will take it from here. We will let you know once we have HTTPS configured and the login flow tested.

Thanks again for your help throughout this process!

Helen Grogan and Team

Helen Grogan

To: Kelly Price

Reply Reply all Forward

Tue 4/7/2026 5:12 PM

Hi Kelly,

Thank you for your help with this. When we inspected the certificate file at /etc/certificate/accounting-game_cse_eng_auburn_edu.crt, it appears to be a Comodo CA root certificate rather than a certificate issued for our server (accounting-game.cse.eng.auburn.edu). This is currently causing a key mismatch error in nginx.

Could you check if the correct server certificate was placed in that directory? We would need a certificate with the subject matching accounting-game.cse.eng.auburn.edu rather than the CA certificate.

Thanks again for your help!

Helen Grogan and Team

Kelly Price

To: Helen Grogan

You replied on Wed 4/8/2026 8:51 AM

accounting-game_cse_eng_a...

3 KB

Is this what you need?

Helen Grogan

To: Kelly Price

Reply

You replied on Wed 4/8/2026 9:19 AM

I am unable to access the file from here, but if you upload that file to the server I can run it against some of our info and check.

Helen Grogan and Team

KP Kelly Price

To: Helen Grogan

You replied on Wed 4/8/2026 10:32 AM

I put it in /etc/certificate/accounting-game_cse_eng_auburn_edu_cert.crt

HG Helen Grogan

To: Kelly Price

It looks like that is the correct certificate, thank you so much for all of your help!

War Eagle!

Helen Grogan and Team

Advisory Board Presentation Invite

GD Gerry Dozier

Reply Reply all Forward

To: Jayden LaCross; Edward Calinsky; Sarah Sessions; Cole Suchan; Helen Grogan

Wed 4/8/2026 9:20 AM

Cc: Andy McLelland

Dear AI for Accounting Games Team:

Congratulations!!!, You are doing an excellent job!!!

Dr. Hari Narayanan, Chair of CSSE, would like for you to present your work to the CSSE Advisory Board on TOMORROW in the McCartney Suite (3rd floor of Shelby Center) @ 2pm.

Would it be possible to present your poster???

We'll have easels set up for you poster 🙂!!!

Your team is so impressive!!!

Can we get this done???

Best Wishes,

gerry

P.S.: If you want to talk, you can contact me at 334-703-2070

Automated Population of Database

To have information to display in our leaderboard and dashboard scenes a script was written to populate our database with sample users and randomly generated information.

populate.py > ...

```python
import psycopg2
import random
import string

# Database connection
DB_CONFIG = {
    "dbname": "leaderboard",
    "user": "postgres",
    "password": "Turmoil4-Mouse8-Attic8-Shorthand4-Catsup8",
    "host": "127.0.0.1",
    "port": "5432"
}

STUDENTS = [
    ("John", "Smith"), ("Jane", "Doe"), ("Aubie", "Tiger"),
    ("Otto", "Priminger"), ("Bruce", "Wayne"), ("Clark", "Kent"),
    ("Barbie", "Doll"), ("The", "Chosen"), ("Shayne", "Topp"),
    ("Bubba", "Gump"), ("Princess", "Buttercup"), ("Indiana", "Jones")
]

GAMES = ["game1", "game2", "game3-1", "game3-2", "game3-3"]
SECTIONS = ["001", "002", "003"]

def generate_username(first, last):
    """
    Format: 1st initial + random letter + 1st initial of last name + 0 + 3 random numbers
    Example: John Smith -> js0123 (with a random letter in the 2nd slot)
    """
    f_init = first[0].lower()
    rand_char = random.choice(string.ascii_lowercase)
    l_init = last[0].lower()
    rand_nums = random.randint(100, 999)
    return f"{f_init}{rand_char}{l_init}0{rand_nums}"

def populate():
    conn = psycopg2.connect(**DB_CONFIG)
    cur = conn.cursor()

    try:
        # 1. Clear existing data
        cur.execute("TRUNCATE TABLE public.game_analytics CASCADE;")
        cur.execute("TRUNCATE TABLE public.player_profiles CASCADE;")

        print("Cleaning out old data and populating new roster...")

        for i, (first, last) in enumerate(STUDENTS):
            # Apply your new username logic
            username = generate_username(first, last)

            # Distribute students across sections
            section = SECTIONS[i % 3]

            # Insert Profile
            cur.execute(
                "INSERT INTO public.player_profiles (username, first_name, last_name, section) VALUES (%s, %s, %s, %s)",
                (username, first, last, section)
            )

            # 2. Generate Scores
            for game in GAMES:
                attempts = random.randint(3, 5)
                for _ in range(attempts):
                    score = random.randint(500, 2500)
                    time_played = random.randint(60, 300)
                    cur.execute(
                        "INSERT INTO public.game_analytics (game, username, score, time_played) VALUES (%s, %s, %s, %s)",
                        (game, username, score, time_played)
                    )

        conn.commit()
        print(f"Successfully added {len(STUDENTS)} students across {len(SECTIONS)} sections.")
        print(f"Sample Username Created: {generate_username('John', 'Smith')}")

    except Exception as e:
        print(f"Error: {e}")
        conn.rollback()
    finally:
        cur.close()
        conn.close()

if __name__ == "__main__":
    populate()
```

Sponsor Approval

AM

Andy McLelland

To: Helen Grogan

Reply Reply all Forward

Fri 4/24/2026 2:47 PM

Helen,

Please consider it approved. Wish I had been able to see and test all the accomplishments, as it came together right at the end of the semester. Thank you and the team for improving the game and the system to track players use.

Best of wishes to you and the team.

"It's a Great Day for Managerial Accounting"

Andrew J. McLelland, PhD
Associate Professor
Harbert College of Business
405 W. Magnolia Ave.
Auburn University, AL 36849-5247
P: 334-844-6217
mclelaj@auburn.edu

Reply Forward